

# SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

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## COURSE OUTCOME COVERED

**CO2: Design Finite Automata DFA, NFA,  $\epsilon$ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)**

*Assessment Tool Weightage: 40%*

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QUESTIONS: 5

Total Marks:25

Duration : 1 Hour

## Design Task

### **Code of Conduct:**

*I Shaik Mohammed Waseem Rayan (192373004) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary*

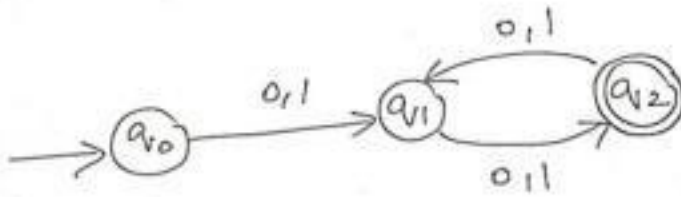
*action.*

**Signature of the student:**

**Shaik Mohammed Waseem Rayan**

# SIMATS ENGINEERING ASSESSMENT TOOL

1. Design the regular expression for the language represented by the given



automata

2. Consider the following

grammar  $S \rightarrow aSc|T|U|e$

$T \rightarrow bTc|e$

$U \rightarrow aUb|e$

A) What is the language defined by the grammar?

B) Give the derivation tree for the word aaabbc

C) A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

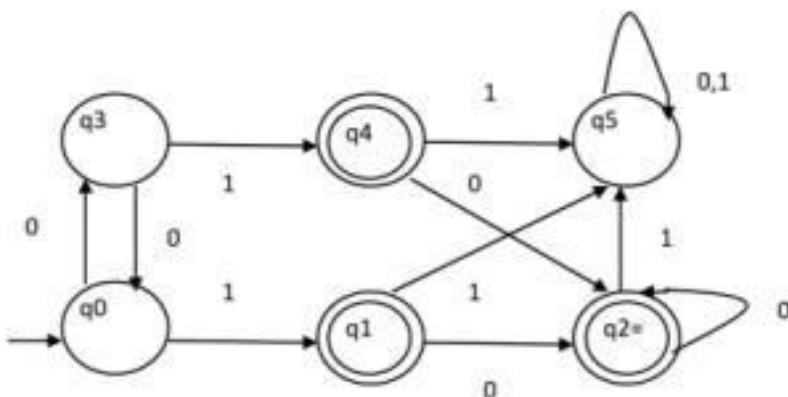
3. Design a non deterministic finite automata with  $\epsilon$ -transitions to represent the language denoted by the regular expression

$00^*1+11^*0$

4. I) Show that if  $L_1$  and  $L_2$  are regular then  $(L_1+L_2)$  also

regular li) Is the language  $\{a^p | p \text{ is prime}\}$  regular? Justify

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states



# SIMATS ENGINEERING ASSESSMENT TOOL

## RUBRICS FOR CALCULATION:

| Criteria                          | Weightage | Level 4 – Exemplary (5 Marks)                                | Level 3 – Proficient (4 Marks)         | Level 2 – Developing (2–3 Marks)             | Level 1 – Beginning (0–1 Mark)  |
|-----------------------------------|-----------|--------------------------------------------------------------|----------------------------------------|----------------------------------------------|---------------------------------|
| <b>Conceptual Understanding</b>   | <b>30</b> | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps     | Basic understanding with some misconceptions | Poor or incorrect understanding |
| <b>Accuracy of Answer</b>         | <b>25</b> | Completely correct answer with no errors                     | Mostly correct with minor errors       | Partially correct with noticeable errors     | Incorrect or irrelevant answer  |
| <b>Application of Knowledge</b>   | <b>20</b> | Correctly applies concepts to solve the question/problem     | Applies concepts with minor mistakes   | Limited or partially correct application     | No or incorrect application     |
| <b>Completeness of Response</b>   | <b>15</b> | Covers all required parts of the question                    | Covers most parts with minor omissions | Covers only some parts                       | Incomplete or missing response  |
| <b>Clarity &amp; Presentation</b> | <b>10</b> | Clear, concise, and well organized answer                    | Understandable with minor issues       | Somewhat unclear or poorly structured        | Difficult to understand         |

**1. Design the Regular Expression for the given automaton Automaton**

- Start State: **q0**
- Final State: **q2**
- $q0 \rightarrow q1$  on **0,1**
- $q1 \leftrightarrow q2$  on **0,1**

The automaton accepts all binary strings of **even length ( $\geq 2$ )**.

**Regular Expression**

$$((0+1)(0+1))^+$$

or equivalently,

$$(0+1)(0+1)((0+1)(0+1))^*$$

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**2. Consider the Grammar**

$$S \rightarrow aSc \mid T \mid U \mid \varepsilon$$

$$T \rightarrow bTc \mid \varepsilon$$

$$U \rightarrow aUb \mid \varepsilon$$

**(A) Language Defined**

The grammar generates

- $a^n c^n$
- $b^n c^n$
- $a^n b^n$

including the empty string.

Hence,

$$L = \{a^n c^n\} \cup \{b^n c^n\} \cup \{a^n b^n\}, n \geq 0$$

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**(B) Derivation for aaabbcc**

Leftmost derivation:

S

$\Rightarrow aSc$

$\Rightarrow aaScc$

$\Rightarrow aaaUcc$

$\Rightarrow aaaUbcc$

$\Rightarrow aaabbcc$

**Parse Tree**

```

  S
 /|\
a S c
 /|\
a S c
 /|\
a U c
```

$/\backslash$   
 $b \cup$   
 $|$   
 $b$   
 $|$   
 $\epsilon$

**(C) Is the grammar ambiguous? Yes.**

Reason:

The string  $\epsilon$  has multiple derivations.

$S \Rightarrow \epsilon$  and

$S \Rightarrow T \Rightarrow \epsilon$

and

$S \Rightarrow U \Rightarrow \epsilon$

Since one string has more than one derivation tree, the grammar is **ambiguous**.

**3. Design an  $\epsilon$ -NFA for**

$$00^*1 + 11^*0$$

This represents

$$0^+1 \cup 1^+0$$

**$\epsilon$ -NFA**

$\epsilon$   
 $\rightarrow q_0 \quad \rightarrow q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_2(F)$   
 $|$   
 $|$   
 $\xrightarrow{\epsilon} q_3 \xrightarrow{1} q_3 \xrightarrow{0} q_4(F)$

Transition Table

| State $\epsilon$ | 0          | 1          |
|------------------|------------|------------|
| $q_0$            | $q_1, q_3$ | -          |
| $q_1$            | -          | $q_1, q_2$ |
| $q_2$            | -          | -          |
| $q_3$            | -          | $q_3$      |
| $q_4$            | -          | -          |

Final states =  **$q_2, q_4$**

**4(i). Show that if  $L_1$  and  $L_2$  are regular then  $L_1 + L_2$  is regular**

Since  $L_1$  and  $L_2$  are regular, there exist DFAs  $M_1$  and  $M_2$ .

Construct an NFA by introducing a new start state with  $\epsilon$ -transitions to  $M_1$  and  $M_2$ .

The NFA accepts

$$L_1 \cup L_2$$

Since every NFA has an equivalent DFA,

$L_1 + L_2$  is regular.

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4(ii). Is

$$L = \{a^p \mid p \text{ is prime}\}$$

regular?

**Answer: No**

Using the Pumping Lemma, Assume L is regular.

Take

$$w = a^p$$

where  $p$  is a prime greater than the pumping length.

After pumping,

$$xy^2z$$

produces a string whose length is no longer prime.

This contradicts the Pumping Lemma.

Hence,

$L = \{a^p \mid p \text{ is prime}\}$  is NOT regular.

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## 5. DFA

### Minimization Initial

#### States

Start :  $q_0$

Final States :

$q_1$

$q_2$

$q_4$

#### Equivalent States

Using partition refinement,

- $q_1 \equiv q_4$

- $q_2 \equiv q_5$

Therefore,

Merge

$$A = \{q_1, q_4\}$$

$$B = \{q_2, q_5\}$$

Remaining states

$q_0$

$q_3$

A={q1,q4}

B={q2,q5}

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### Minimized DFA

#### State 0 1 Final

q0    q3 A No

q3    q0 A No

A    B B Yes

B    B B Yes

#### Initial State

→ q0

#### Final States

\* A={q1,q4}

\* B={q2,q5}

#### Minimized DFA Diagram (Text)

